

Practical 8 - Matched studies and their analysis

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

Datasets required: `vit_E.csv`

Introduction

The aim of this practical is to consolidate the ideas of matching discussed in the lecture. We will use data from a matched case-control study of the association between vitamin E measured in the blood and occurrence of cancer. The dataset is called `vit_E.csv`.

High levels of vitamin E are thought by some to be protective against cancer. This hypothesis was investigated by measuring Vitamin E in stored blood samples from 271 men in a large cohort study initially aged 35–64 years who subsequently developed cancer. These values were compared with those from control men who had not, at that time, developed cancer. One control for each case was selected randomly from within the cohort study, subject to matching for age (within 5 years), duration of storage of the blood sample (within 3 months) and smoking status.

Note that this dataset does not include the matching variables. It does include ‘observation time’, which is the time from blood collection to diagnosis of cancer in the cases. In this session we will analyse the data assuming no confounding.

Variable	Description
<code>set</code>	Case-control unique identifier taking values from 1 to 271
<code>case</code>	0: control, 1: case
<code>vitE</code>	Vitamin E (mg/dl)
<code>time</code>	Observation time: 1: up to 1 year, 2: 1–3 years, 3: over 3 years

Aims

- Understand how to appropriately conduct hypothesis tests on data from a matched case-control study.
- Use appropriate statistical tests to analyse matched studies with continuous dependent variables.

Analysis

Question 1

Load the data and take a look at the first few rows to ensure you understand how it is structured.

Question 2

Summarize the vitamin E measurements for cases and controls. What is the difference between the means in the two groups? Use suitable plots to display vitamin E levels in the cases and controls.

Question 3

For questions 3-5 we will analyse these data treating vitamin E as the dependent variable, and cancer as the independent variable. In reality the vitamin E reading is best thought of as an exposure since blood was taken and stored before the onset of cancer.

We will first analyse these data using linear regression.

We first demonstrate an invalid approach, to demonstrate the importance of allowing for block in any analysis. Fit a linear regression model with just the case-control variable as an explanatory variable.

```
lm1 <- lm( vitE ~ case, data=vit)
summary(lm1)
```

Now add the set variable, as a factor covariate in the regression model.

```
lm2 <- lm( vitE ~ case + factor(set), data=vit)
summary(lm2)
```

Compare the estimates of the case-control coefficient and its standard error from the two analyses. Is there evidence that people who developed cancer had different vitamin E levels than people who did not develop cancer?

Question 4

In order to make the data suitable for analysis with a paired t-test (and other analyses), we need to reconfigure it so that the data from each pair is contained in one row. This can be done using the reshape function, which is an extremely useful command, but can be fiddly to get to do what you want it to do.

Use the following line to reshape the data to wide form:

```
vitWide <- reshape(vit, v.names=c("vitE"), timevar="case",  
                    idvar="set",direction="wide")
```

Then use `head` to check how `vitWide` looks.

Question 5

Using the data in wide format, carry out a paired t-test of the null hypothesis that the mean blood vitamin E levels are the same in cases and controls.

How do the estimated mean difference and its standard error compare to those from the linear regression models above?

Carry out some assessment of the assumptions made by the paired t-test.

Question 6

For the remainder of this session we will dichotomise vitamin E into a binary variable and perform analysis to estimate the odds ratio of association. Here we can use the reversible nature of an odds ratios so that we can interpret vitamin E as the exposure and cancer as the outcome or vice-versa.

In `vitWide`, create variables `vitECaseHigh` and `vitEControlHigh` that indicate whether the case (respectively control) have a vitamin E level above 12mg/dl. Then cross-tabulate these two variables.

Question 7

Using the table generated in the previous question, perform McNemar's test using `binom.test`.

Question 8

Use the table you generated in question 6 and the output from `binom.test` in question 7 to estimate the odds ratio and a 95% CI for it.

Question 9

What are your epidemiological conclusions from the analysis where vitamin E status is dichotomised? Working together with one or more colleagues, write a short paragraph to summarise your findings.